Aircraft RUL Prediction — Presentation Revision

1. Predictive Maintenance

Using machine/sensor data to predict when a machine may fail or need maintenance.
Say: “Our project uses predictive maintenance to estimate the remaining useful life of an aircraft engine before failure.”

2. RUL — Remaining Useful Life

Estimated number of operating cycles remaining before an engine reaches the end of its useful life.
Target: RUL = maximum cycle − current cycle.

3. NASA C-MAPSS

NASA's Commercial Modular Aero-Propulsion System Simulation dataset containing simulated aircraft-engine sensor data. **Our dataset:** FD001.

4. Time-Series Data

Sensor readings are recorded over engine operating cycles, so the order of observations matters.

5. EDA — Exploratory Data Analysis

Understanding the data before modeling: columns, distributions, missing values, constant features, and sensor trends/correlations.

6. Feature Engineering

Creating useful model inputs from original sensor data.
Our features: cycle-to-cycle delta, rolling mean (5), rolling mean (10), rolling standard deviation (10).

7. Delta

Difference between the current sensor reading and the previous reading. Captures short-term change/trend.

8. Rolling Mean

Average of recent sensor readings over a fixed window. Smooths noise and captures local trend.

9. Rolling Standard Deviation

Measures variation in recent sensor readings and helps capture changes in sensor stability.

10. XGBoost

Full form: Extreme Gradient Boosting.
A tree-based supervised ML algorithm that builds trees sequentially, with later trees correcting previous errors.
Why: effective for tabular data and nonlinear relationships.

11. Regression

RUL is a continuous numerical value, so this is a regression problem, not classification.

12. Feature Importance

Shows which input features contributed most to the model's predictions. The top feature was **cycle**; important engineered features included rolling means of sensors 4, 11 and 15.

13. MAE

Mean Absolute Error: average absolute difference between predicted and actual RUL. Lower is better; easy to interpret in cycles.

14. RMSE

Root Mean Squared Error: square root of average squared error. Lower is better and large errors are penalized more.

15. R^2

R-squared measures how much variance in the target is explained by the model. Higher is better. **Do not call R^2 accuracy.**

Results to Memorize

Item	Value
Features	65
Algorithm	XGBoost Regression
Validation MAE	23.58 cycles
Validation RMSE	30.97 cycles
Validation R^2	0.778
Test MAE	19.71 cycles
Test RMSE	27.68 cycles
Test R^2	0.556

Project Pipeline

NASA C-MAPSS → Preprocessing → EDA → Feature Engineering → 65 Features → XGBoost Regression → Evaluation → Saved Model (.pkl) → Streamlit → CSV Upload → RUL Prediction → Health Status

Short Presentation Answer

“We built an aircraft-engine predictive maintenance system using NASA C-MAPSS FD001 data. We calculated RUL from engine cycles and performed EDA to identify useful sensors. We engineered delta, rolling mean and rolling standard deviation features, giving 65 model features. We trained XGBoost regression because RUL is continuous and XGBoost handles nonlinear tabular relationships well. Finally, we saved the model and deployed it through Streamlit for RUL prediction and health status.”

Important Tip

For $R^2 = 0.556$, say: “The model explains about 55.6% of the variance in the test RUL values.” Do not say “55.6% accuracy.”